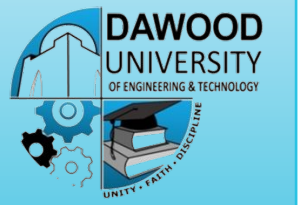
LECTURE – SUPER
OOP (JAVA)



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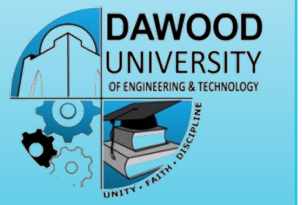
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AGENDA

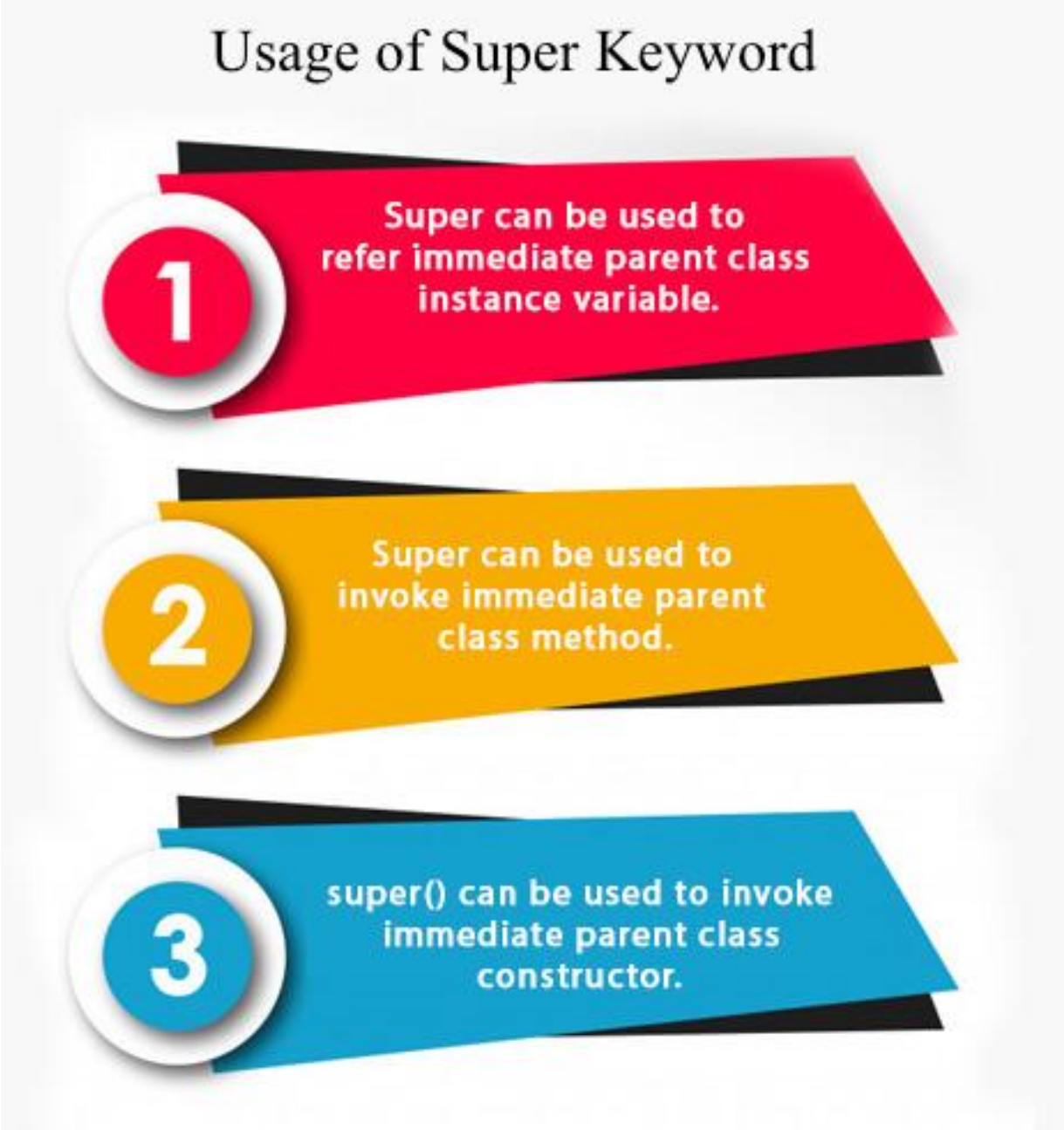


- INTRODUCTION TO SUPER KEYWORD
- USAGE OF SUPER KEYWORD
- FINAL KEYWORD

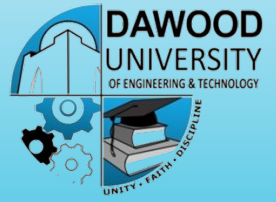
SUPER KEYWORD



- The **super** keyword in Java is a reference variable which is used to refer immediate parent class object.
- Whenever you create the instance of subclass, an instance of parent class is created implicitly which is referred by super reference variable.


Usage of Super Keyword
1
Super can be used to refer immediate parent class instance variable.
2
Super can be used to invoke immediate parent class method.
3
super() can be used to invoke immediate parent class constructor.

SUPER IS USED TO REFER IMMEDIATE PARENT CLASS INSTANCE VARIABLE.

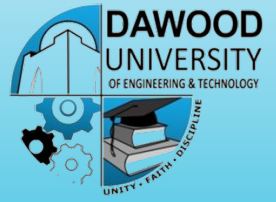



```
class Animal{
String color="white";
}
class Dog extends Animal{
String color="black";
void printColor(){
System.out.println(color);//prints color of Dog class
System.out.println(super.color);//prints color of Anima
l class
}
}
class TestSuper1{
public static void main(String args[]){
Dog d=new Dog();
d.printColor();
}}
```

2) SUPER CAN BE USED TO INVOKE PARENT CLASS METHOD

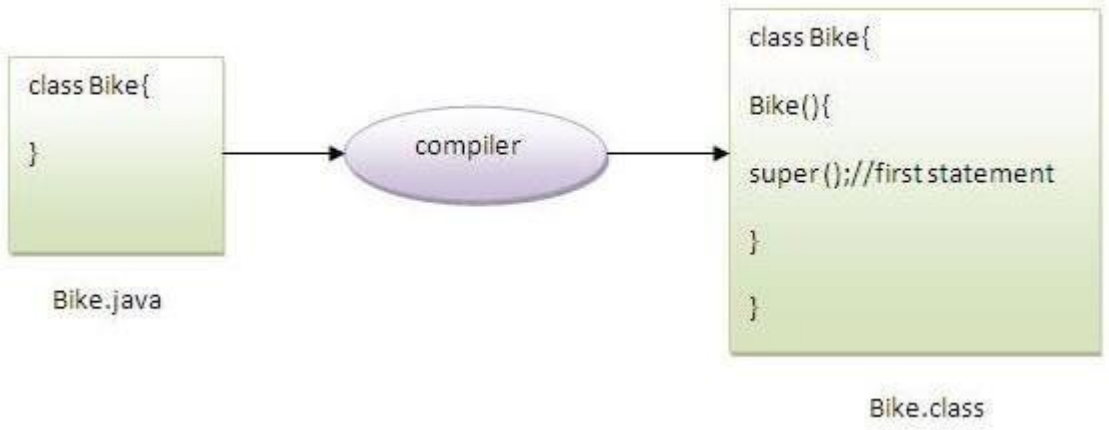
```
class Animal{
void eat(){System.out.println("eating...");}
}
class Dog extends Animal{
void eat(){System.out.println("eating bread...");}
void bark(){System.out.println("barking...");}
void work(){
super.eat();
bark();
}
}
class TestSuper2{
public static void main(String args[]){
Dog d=new Dog();
d.work();
}}
```

3) SUPER IS USED TO INVOKE PARENT CLASS CONSTRUCTOR.




```
class Animal{
Animal(){System.out.println("animal is created");}
}
class Dog extends Animal{
Dog(){
super();
System.out.println("dog is created");
}
}
class TestSuper3{
public static void main(String args[]){
Dog d=new Dog();
}}
```

- As we know well that default constructor is provided by compiler automatically if there is no constructor. But, it also adds super() as the first statement.

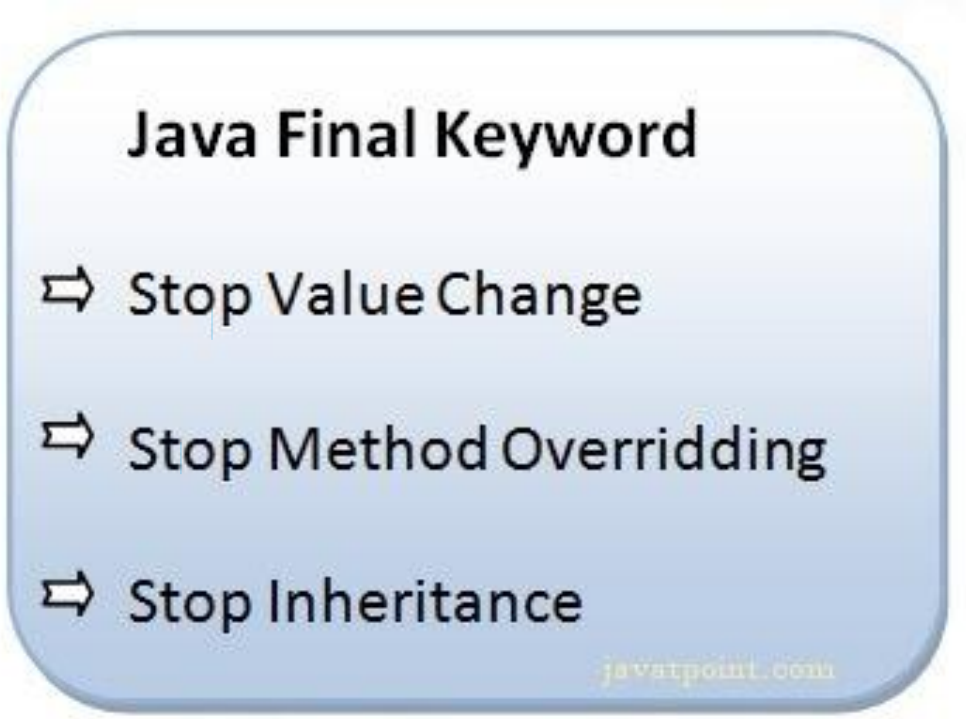


- **Note: super() is added in each class constructor automatically by compiler if there is no super() or this().**

FINAL KEYWORD

- The **final keyword** in java is used to restrict the user. The java final keyword can be used in many context. Final can be:

1. variable
2. method
3. class



FINAL VARIABLE IN JAVA

- If you make any variable as final, you cannot change the value of final variable(It will be constant).
- There is a final variable speedlimit, we are going to change the value of this variable, but It can't be changed because final variable once assigned a value can never be changed.

```
class Bike9{
 final int speedlimit=90;//final variable
 void run(){
  speedlimit=400;
 }
 public static void main(String args[]){
 Bike9 obj=new  Bike9();
 obj.run();
 }
}
```

2) JAVA FINAL METHOD

- If you make any method as final, you cannot override it.

```
class Bike{
  final void run(){System.out.println("running");}
}

class Honda extends Bike{
   void run(){System.out.println("running safely with 100kmph");}

   public static void main(String args[]){
   Honda honda= new Honda();
   honda.run();
   }
}
```

Output:Compile Time Error

JAVA FINAL CLASS

- If you make any class as final, you cannot extend it.

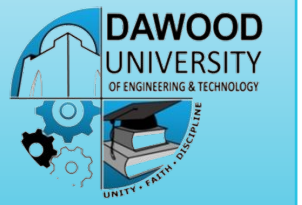
```
final class Bike{}

class Honda1 extends Bike{
  void run(){System.out.println("running safely with 10
0kmph");}

  public static void main(String args[]){
  Honda1 honda= new Honda1();
  honda.run();
  }
}
```

Output:Compile Time Error

SOME IMPORTANT QUESTIONS



- **Q) Is final method inherited?**
- Can we declare a constructor final?





Thanks!

For The Time & Attention

Do You Have Any Question?

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